Safety-Monitored Edge Perception for ADAS: Connecting Runtime Monitoring Evidence to a Bounded Safety Argument

Pruthu Parikh — draft v0.9 (Week 9). IEEE-style structure; PDF build in Week 10. All numbers generated by `scripts/build_report_assets.py` from `results/*` (see `paper/tables.md`); no hand-typed metrics.

Abstract

Deep-learning perception is the least certifiable component of an ADAS stack: it fails silently, out of its training distribution, without an error signal. This work builds a camera-only pedestrian/vehicle/cyclist detector for an AEB-style function and — rather than maximizing accuracy — asks the safety-engineering question: *when should this stack not be trusted, and can that be detected at runtime within a real-time budget?* We train YOLOv8n on KITTI (val mAP50 0.859), deploy via TensorRT FP16 (mAP50 delta –0.0024, pure-inference 2.6 ms), calibrate confidences by temperature scaling (ECE 0.081 → 0.039), and gate the output with a frame-level monitor: a max-confidence OOD score with Q95/Q99 thresholds frozen from validation data drives a three-state machine (`NOMINAL`/`DEGRADED`/`FAIL_SAFE_REQUEST`) triggering only on sustained breaches. On BDD100K shift slices the score separates night from in-distribution data at AUROC 0.982; at the frozen Q95 threshold the monitor retains 94.8% of in-distribution frames while accepting only 6% of night frames. Under synthetic fault injection the monitor flags 99% of frames as severe fog collapses mAP50 from 0.847 to 0.071, with the full perception+monitor loop at p95 17.2 ms against a 40 ms budget. Equally important, the campaign quantifies a blind spot: severe low-light removes 19% of mAP50 while the monitor stays 97.7% `NOMINAL`. We package both results — capability and documented residual — into a GSN safety case, SOTIF argument, and ISO/PAS 8800-aligned traceability from hazards (STPA/HARA) to per-requirement evidence. The contribution is methodological: an end-to-end, reproducible template connecting edge-ML performance artifacts to safety-engineering evidence, with claims bounded to a defined ODD and honest residual risks.

I. Introduction

Advanced driver-assistance functions such as automatic emergency braking (AEB) place a neural detector in a safety-critical control loop. Two properties make this uncomfortable for classical functional-safety practice: the detector's failure modes are input-dependent rather than component-fault-driven, and its confidence outputs are not calibrated probabilities. ISO 21448 (SOTIF) frames this as *performance limitations under triggering conditions*; ISO/PAS 8800 extends the lifecycle view to AI-specific evidence. What is often missing in portfolio and research prototypes is the connective tissue: metrics exist, safety documents exist, but nothing traces a specific AUROC number to a specific hazard.

This project builds that connective tissue for a deliberately narrow item: camera-only 2D detection of pedestrians, vehicles, and cyclists within a KITTI-like operational design domain (ODD), supervised by a runtime monitor that requests degraded or fail-safe handling when perception evidence degrades. Contributions:

1. **A monitored perception MVP on edge hardware** (RTX 3050 Ti, 4 GB): YOLOv8n + TensorRT FP16, calibrated confidence, frame-level OOD score, three-state gating, per-frame evidence logging — p95 17.2 ms full loop.

2. **Threshold and evaluation hygiene as a first-class design rule**: thresholds from validation quantiles only; test and shift slices report-only; seeds and split hashes committed.

3. **A two-campaign triggering-condition exploration** (real BDD100K night/rain/fog slices; deterministic synthetic corruptions) showing where frame-level max-confidence monitoring works — and quantifying where it does not.

4. **A bounded safety argument**: STPA/HARA-derived requirements SR-01..SR-06, each verified by scripted evidence; GSN case with an explicit residual-risk node; SOTIF classification moving the discovered low-light blind spot from unknown-unsafe to known-unsafe.

We claim alignment with safety-engineering practice, not compliance or certified safety.

II. Related Work

Runtime monitoring of DNN perception. Max-softmax/max-confidence baselines (Hendrycks & Gimpel), energy-based OOD scores (Liu et al.), and Mahalanobis feature-space detectors (Lee et al.) define the standard spectrum from output-level to representation-level monitors. We deploy the simplest (max-confidence) after empirically finding our post-NMS energy proxy no better (mean OOD AUROC 0.913 vs 0.928), and carry feature-space methods as future work — an explicit scope decision recorded before evaluation on test data.

Calibration. Temperature scaling (Guo et al.) remains the standard post-hoc method; we apply a scalar T in logit space to detection confidences with correctness defined by class-matched IoU ≥ 0.5 greedy assignment.

Safety engineering for ML perception. STPA (Leveson) supplies unsafe-control-action analysis suited to software-intensive systems; SOTIF (ISO 21448) frames triggering conditions; GSN (SCSC) structures the argument; ISO/PAS 8800 names AI-lifecycle evidence obligations. Our use is educational-depth alignment on a real evidence base.

Robustness benchmarks. Corruption benchmarks (Hendrycks & Dietterich; KITTI-C/CODA variants) motivate our synthetic campaign; we use lightweight deterministic OpenCV corruptions rather than benchmark suites to keep the dependency stack frozen and every image bit-reproducible from (frame-id, corruption, severity).

III. System Design

Item and ODD. Forward camera, KITTI-like mounting; three classes; daylight, clear/lightly overcast, KITTI-style road scenes. Night, adverse weather, and degraded imaging are *outside* the ODD — they are precisely what the monitor targets. Downstream planner/AEB and actuators are simulated boundaries; assumption A-03 (planner honors monitor states) is untested by design and carried as the largest interface risk.

Architecture. `camera → YOLOv8n (PyTorch | TensorRT FP16) → runtime monitor → {gated detections, monitor state} → planner boundary`, with a per-frame machine-readable log as the evidence spine. Module boundaries in code mirror this: `src/dataset`, `src/monitor/{calibration,scoring,thresholds,state_machine,runtime}`, `scripts/*` as the only metric-producing entry points.

Safety derivation. HARA-lite works one hazardous event fully (undetected pedestrian in ego path, urban daytime: S3/E4/C3 → ASIL D as an educational classification) yielding safety goal SG-01; STPA over two control actions (gated detections, monitor state) yields 9 unsafe control actions and 9 causal scenarios, which decompose SG-01 into six measurable requirements: calibrated confidence (SR-01), runtime OOD scoring (SR-02), sustained-breach degraded entry (SR-03), fail-safe request (SR-04), per-frame logging (SR-05), 40 ms p95 latency budget (SR-06).

IV. Monitor Design

Calibration (SR-01). kitti-val (1122 frames) splits 50/50 (seed 42) into calibration-fit and calibration-report subsets. Detections at conf ≥ 0.05 match GT greedily (same class, IoU ≥ 0.5, one-to-one, confidence-descending). A scalar temperature minimizing binary NLL on the fit subset (golden-section search) gives T = 0.600 — the model is *underconfident*; ECE on the report subset improves 0.0812 → 0.0390 (15 equal-width bins).

OOD score (SR-02). Frame-level `max_conf_score = 1 − max(conf)`; empty frames score 1.0 (no evidence of in-distribution input — empirically the dominant failure signature under severe degradation). A post-NMS energy-style proxy (–logsumexp over top-10 confidence logits) is evaluated as the committed comparison and found not materially better; the simpler score is selected by pre-registered policy (energy must win by ≥ 0.01 mean AUROC).

Thresholds (SR-03/04). Q95 = 0.1862 and Q99 = 0.3728: quantiles of the ID score distribution on the calibration-report subset only. BDD and kitti-test never touch threshold selection.

Gating. Sustained-breach state machine: 3 consecutive Q95 breaches enter `DEGRADED`; 2 consecutive Q99 breaches or 10 unrecovered degraded frames enter `FAIL_SAFE_REQUEST`; 5 consecutive clean frames recover one level; breach is strict (`score > threshold`). Single-frame spikes cannot transition — verified deterministically.

V. Experimental Evaluation

All experiments: seed 42, committed split hashes, commands in `docs/experiment_log.md` (EXP-001..EXP-011); tables in `paper/tables.md`.

A. Baseline (Table 1). YOLOv8n, 50 epochs: val mAP50 0.8588 (pedestrian 0.749, vehicle 0.948, cyclist 0.880), mAP50-95 0.556. TensorRT FP16: mAP50 0.8564 (Δ –0.0024), pure engine inference 2.6 ms/img; end-to-end p50 16.9 ms.

B. Calibration (Table 2). ECE 0.0812 → 0.0390 on the disjoint report subset; reliability diagrams confirm the pre-scaling accuracy-above-diagonal signature of underconfidence.

C. OOD separation (Table 3). Against the ID report subset, max-conf AUROC/FPR@95: night 0.982/0.112; rain 0.876/0.537; fog 0.926/0.419 (n=13); clear-day transfer control 0.758/0.624. The transfer control matters: even in-ODD-like BDD imagery is detectably shifted from KITTI, so thresholds cannot transfer across camera domains without re-derivation.

D. Risk-coverage (Table 4). At frozen Q95 the monitor keeps 94.8% of ID frames (accepted-frame mAP50 0.854 vs 0.850 at full coverage) while accepting only 6.0% of night, 36.0% of rain, 23.1% of fog, and 68.0% of clear-day-control frames. The monitor's value is out-of-ODD rejection, not ID accuracy gain — accepted-frame mAP is nearly flat in coverage, and we report that honestly.

E. Runtime and latency (Table 5). Full loop (predict + score + state machine + log row): TRT FP16 p50 14.6 / p95 17.2 ms (67.9 FPS); PyTorch p50 15.7 / p95 17.5 ms. Monitor overhead is within run-to-run variance of the detector alone. 300 ID frames produce zero false transitions (98.3% `NOMINAL`, brief single-frame breaches contained by the sustained-breach rule); gating scenario replay passes 7/7; log completeness 15/15 checks.

F. Fault injection (Table 6). 16 conditions (clean + 5 corruptions × 3 severities) on a seeded 300-frame kitti-test subset, report-only. The monitor tracks abrupt degradation: fog:high collapses mAP50 0.847 → 0.071 and the monitor flags 99% of frames non-`NOMINAL`; blur:high and noise:high reach 78%/77% `FAIL_SAFE_REQUEST` as mAP halves or worse. **The campaign's most important result is negative:** low_light:high removes 19% of mAP50 (0.847 → 0.689) while the monitor stays 97.7% `NOMINAL`; dead_pixels:medium behaves similarly. Mechanism: max-confidence sees the *strongest surviving* detection, not the *missing* ones — gradual recall erosion is invisible to it. All 16 conditions stay within the latency budget (max p95 31.2 ms).

VI. Safety Case and SOTIF Argument

The GSN case (`safety/gsn.svg`) argues the bounded top claim — *evidence-backed detection of degraded perception conditions and fail-safe requests within the 40 ms budget, within the defined ODD and stated assumptions* — through five sub-goals (hazard-driven requirements; measured baseline; monitor effectiveness; latency; verified traceability), each closed by scripted evidence, with an explicit residual node. The SOTIF argument classifies operating regions: in-ODD operation and abrupt severe degradation are argued acceptably handled (known-safe / known-unsafe-mitigated); gradual low-signal degradation is *known-unsafe-unmitigated* — discovered and quantified by EXP-010, previously unknown-unsafe; unexplored real-world conditions remain unknown-unsafe by definition. Traceability is machine-readable end-to-end: `requirements.csv` (derivations), `traceability_matrix.csv` (SR → implementation → test → metric → result → evidence, all six `verified`), `evidence_index.csv` (24 claims → artifacts → experiments).

VII. Limitations

(1) Ratings in HARA-lite are illustrative, not field-calibrated; one hazard worked fully. (2) The planner/actuator boundary is assumed (A-03), never exercised — fail-safe is a *request*. (3) Corruptions are plausibility models (fog = uniform haze), not weather physics; BDD fog n=13. (4) The low-light blind spot is documented, not fixed; candidate mitigations (detection-count plausibility, temporal consistency, Mahalanobis) were pre-registered scope cuts. (5) Thresholds are KITTI-val-specific; the clear-day control shows they do not transfer across camera domains. (6) INT8 quantization was deferred; FP16 sufficed for the budget. (7) Single dataset family in-ODD; single hardware target.

VIII. Conclusion

A small detector, a simple monitor, and disciplined evidence hygiene are enough to build a genuine, bounded safety argument — including the part most prototypes omit: a quantified statement of where the monitor is blind. The template (hazard analysis → measurable requirements → frozen-threshold runtime monitor → scripted verification → GSN/SOTIF packaging, all seeded and traceable) is the transferable artifact; the low-light result is the demonstration that the template surfaces real residual risk instead of hiding it.

References

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